DS-A2 Phase 1
Statistical Inference and Uncertainty

NOAA SanctSound SB02_13 | Observed deployment

Research question
Is the average clock-defined nighttime published hourly dB indicator different from the daytime indicator, and how uncertain is that difference? This is an observational contrast, not a causal effect.

Estimand
Mean across eligible local dates of (night mean hourly dB − day mean hourly dB). Arithmetic means of dB indicators are not linear-energy means.

Design
America/New_York; day 09:00–17:00; night 21:00–05:00 next date; >=6/8 valid hours in both windows. Two-sided alpha=.05. Operational interpretation thresholds: 1 and 3 dB.

Observed result
n=64; mean difference=0.352 dB (median 0.635); IID t 95% CI [-0.428, 1.132] dB, p=0.370. HAC and all MBB sensitivity intervals also include zero. The estimate and interval do not establish a day–night difference; the interval remains compatible with small positive and negative values.

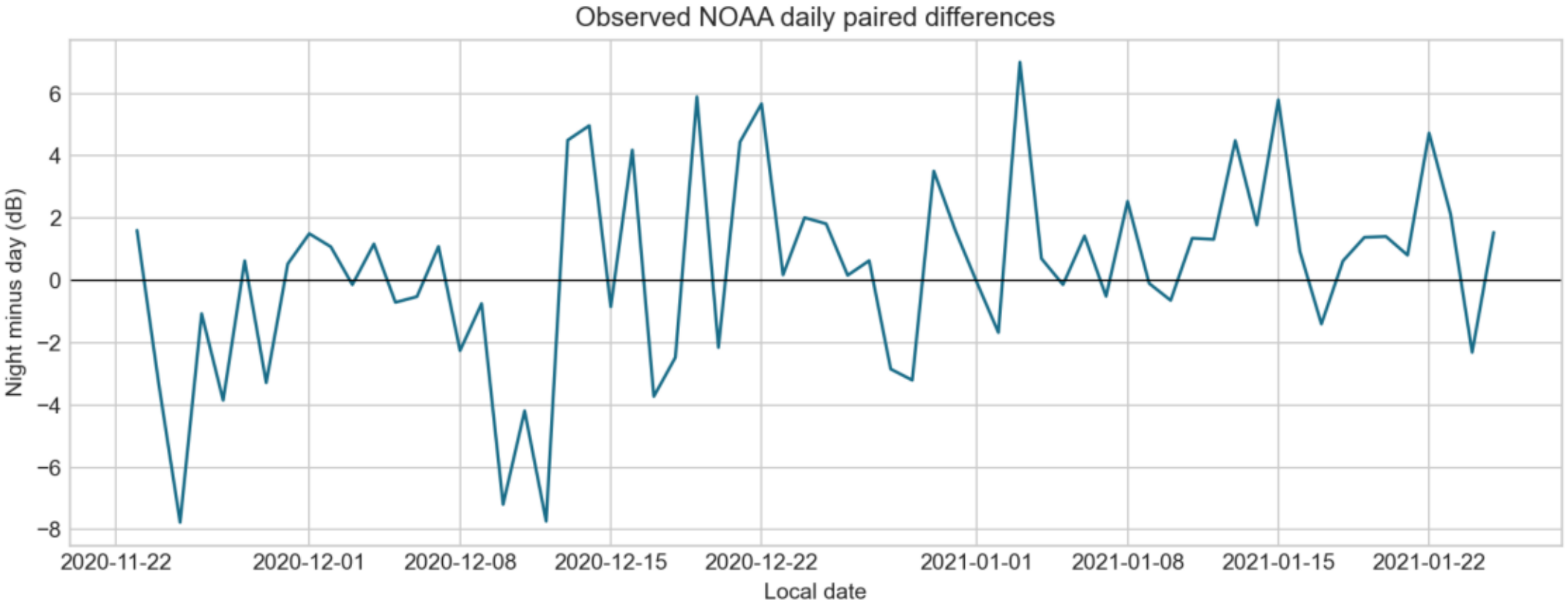
IID power simulation
At n=30, sigma=sigma0=3.123 dB, effect=3 dB, analytical power=0.999 and Monte Carlo rejection=0.999 (5,000 replications). At twice sigma0, rejection=0.713. These are conditional IID model results.

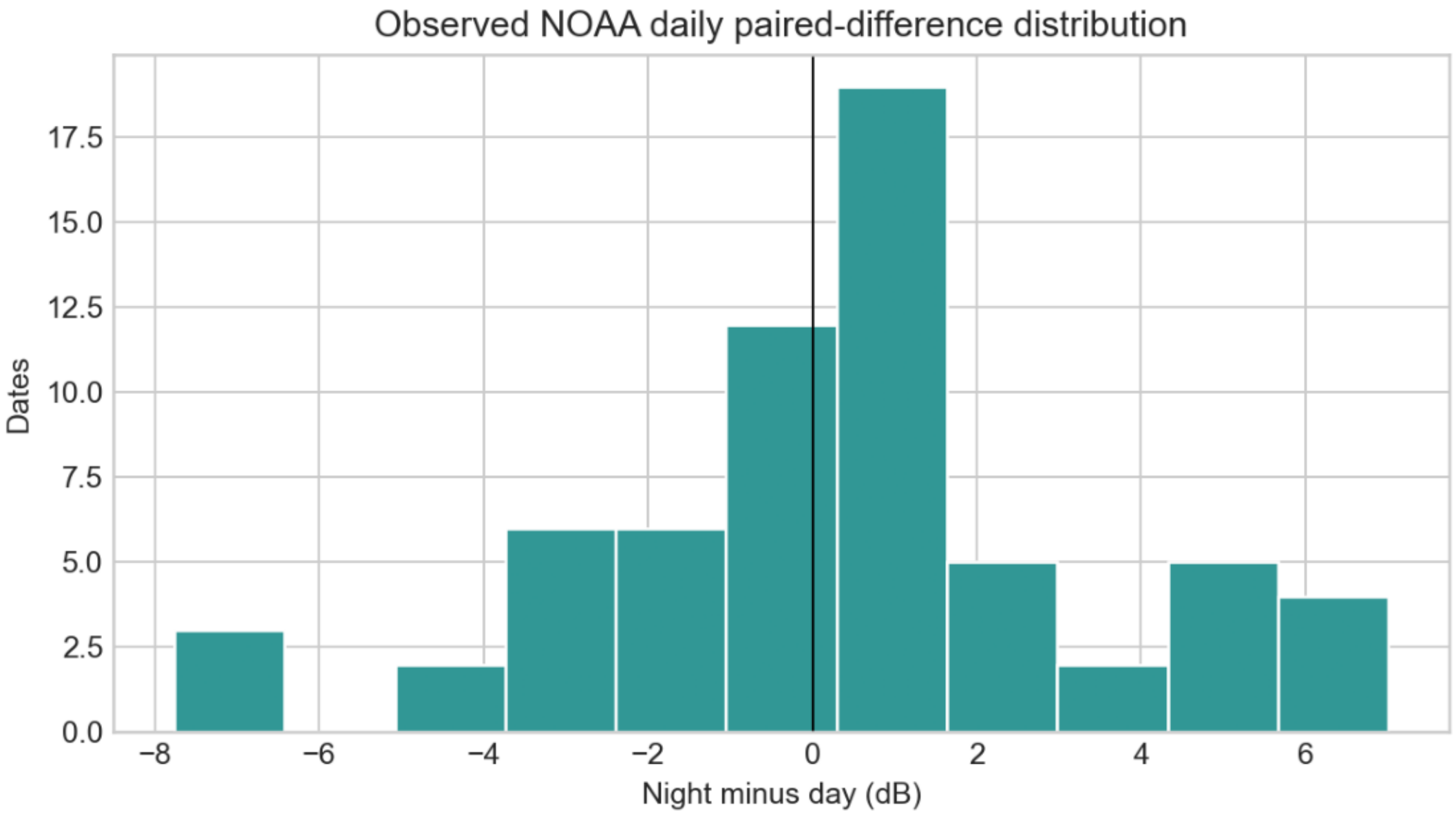
AR(1) counterexample
At rho=.6 and true effect 0, rejection/coverage were IID t 0.327/0.673, HAC 0.155/0.845, and MBB 0.174/0.826 across 2,000 simulated series. HAC and MBB reduce, but do not eliminate, size distortion in this design.

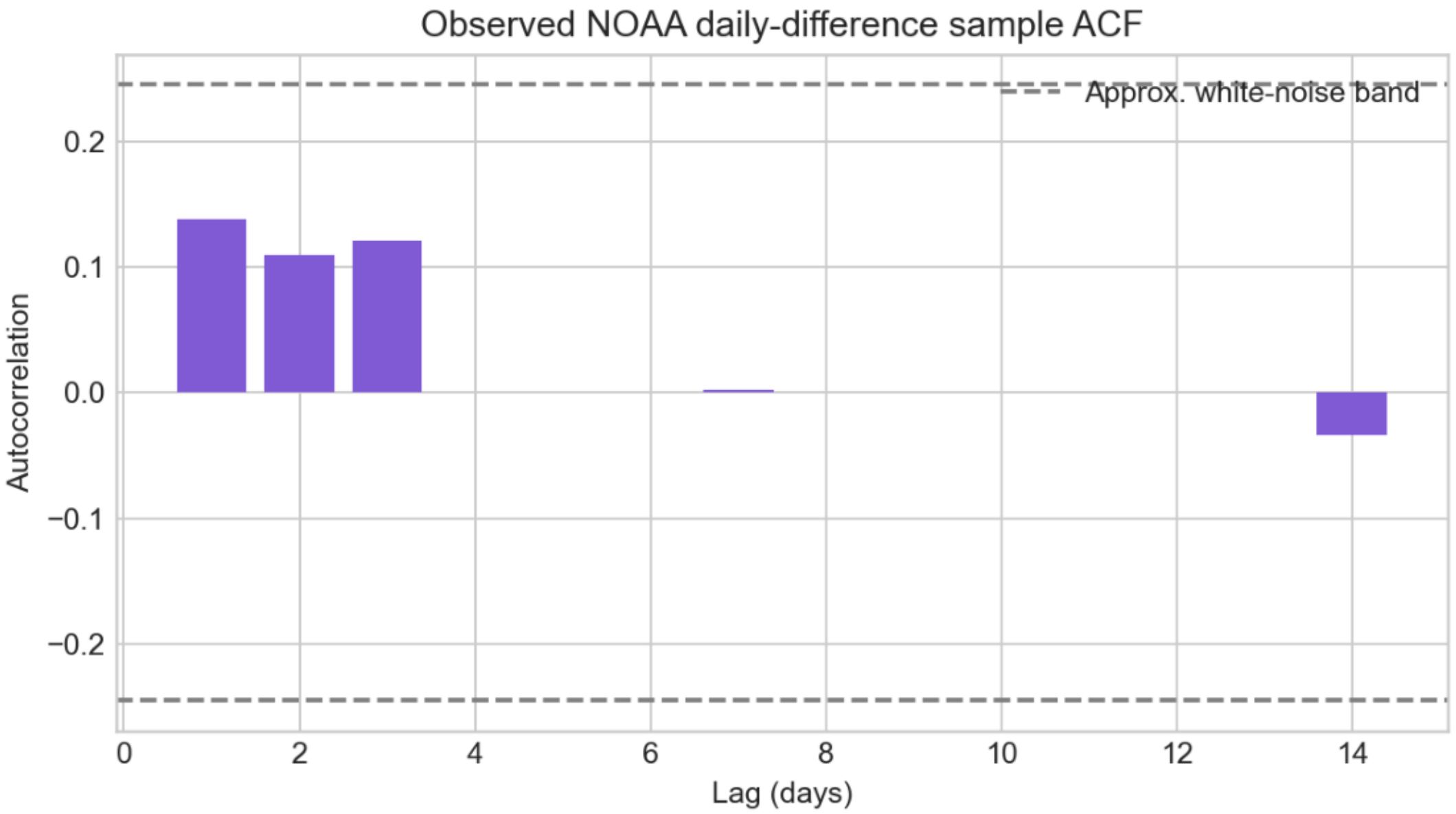
Methods
IID one-sample t baseline; Newey–West Bartlett HAC with frozen bandwidth; overlapping non-circular moving block bootstrap. IID Gaussian sampling/power simulations use empirical marginal SD. Stationary Gaussian AR(1) simulations challenge independence.

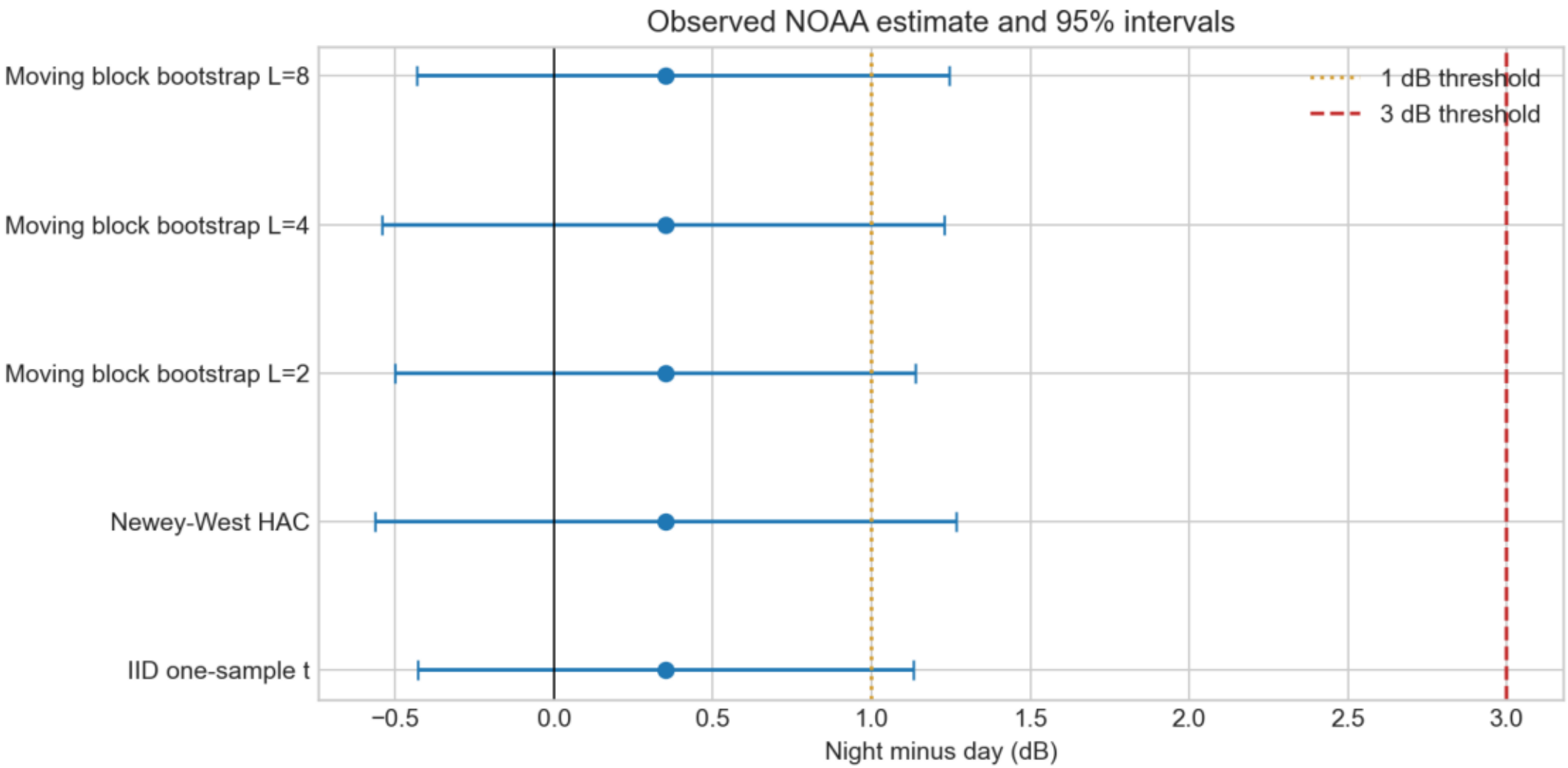
Limitations
64 eligible dates from one short deployment; row-level QC flags absent; lower frequency edge conflicts between the brief JSON (20 Hz) and netCDF/NCEI (10 Hz). Dependence and nonstationarity remain important.

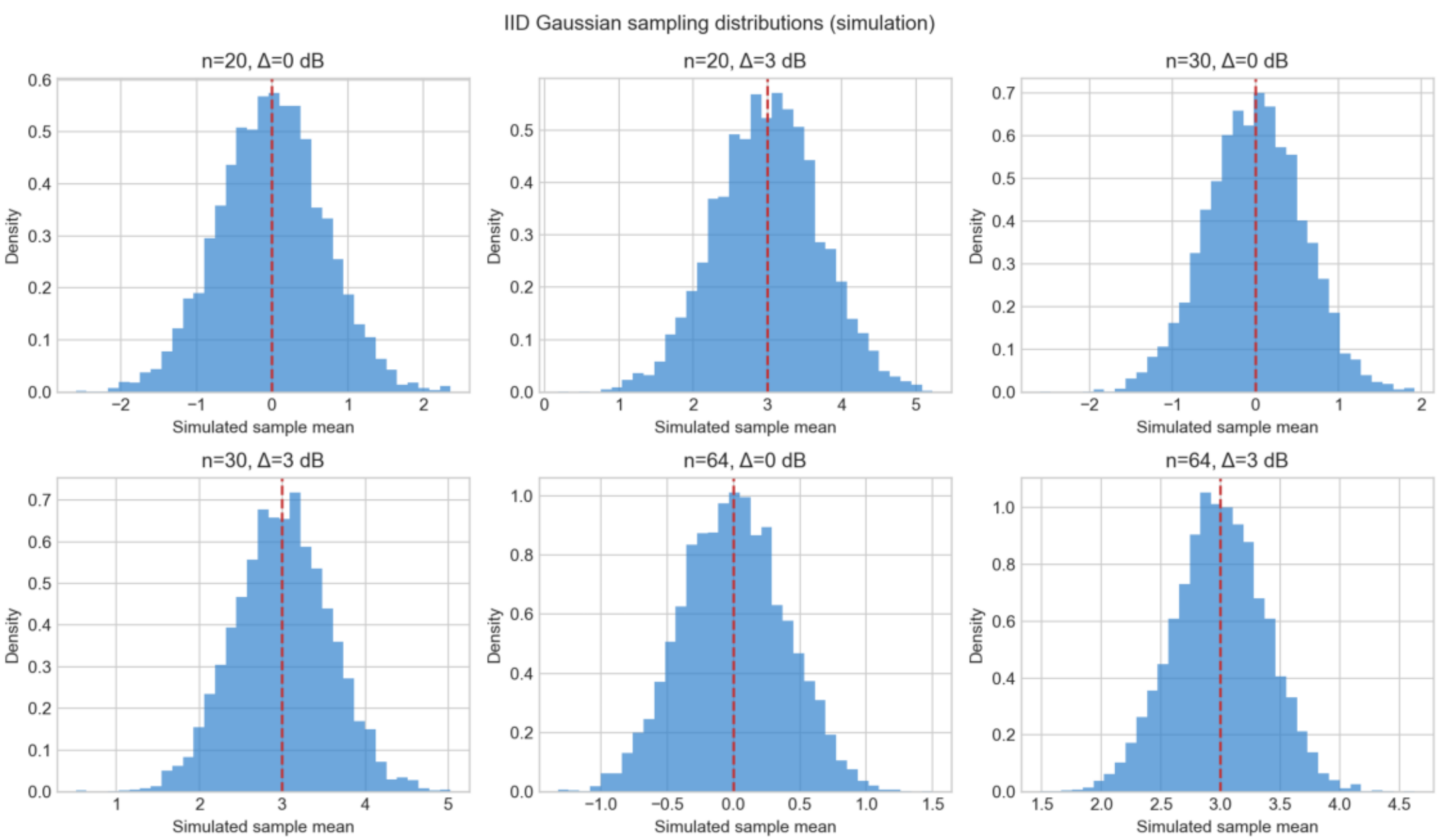
Automated execution record
See the executable notebook, machine-readable results, environment manifest, tests, and AI-use log. No independent human verification is claimed.

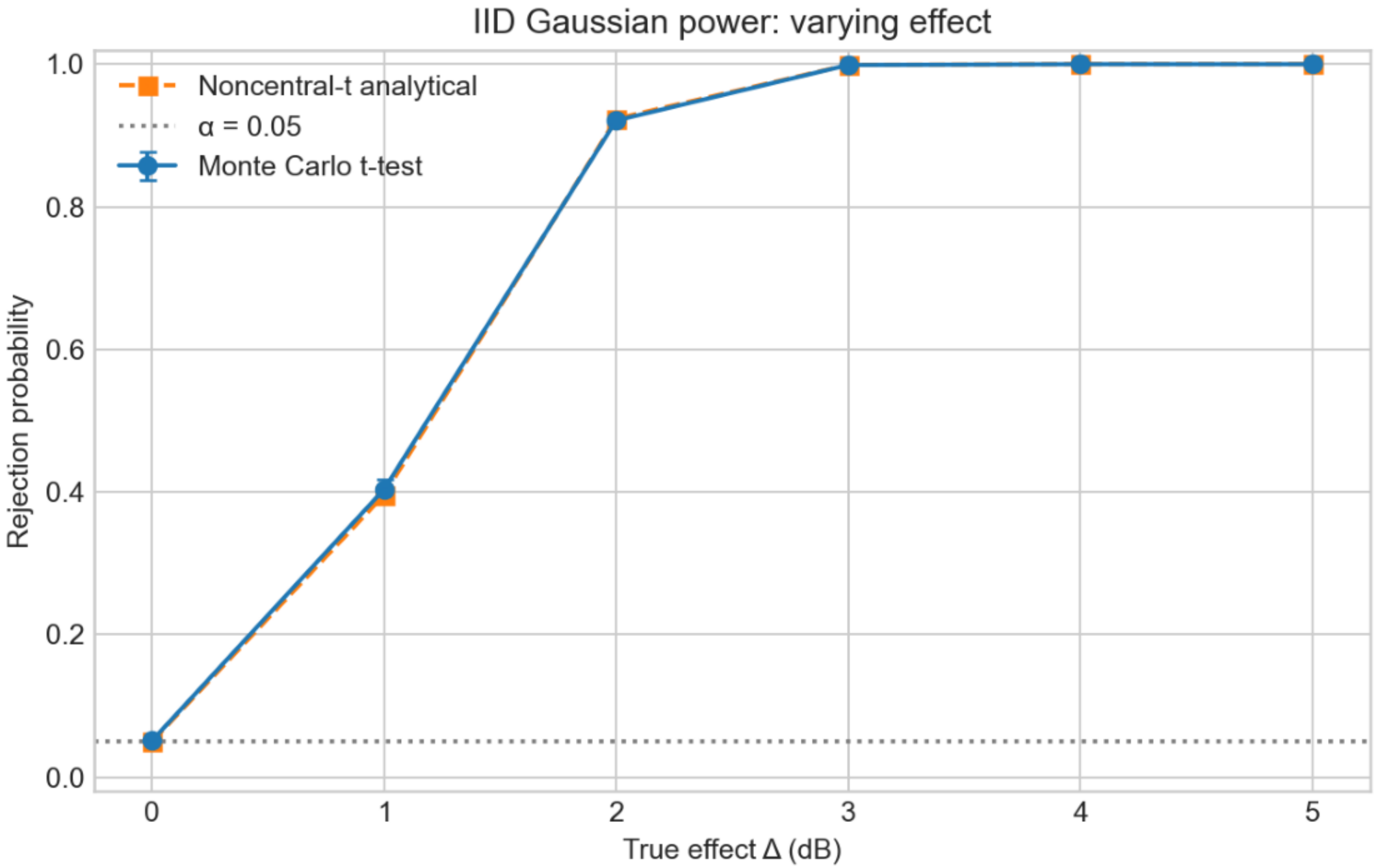

Observed NOAA daily paired differences
Night minus day (dB)
6
4
2
0
−2
−4
−6
−8
2020-11-22
2020-12-01
2020-12-08
2020-12-15
2020-12-22
2021-01-01
2021-01-08
2021-01-15
2021-01-22
Local date

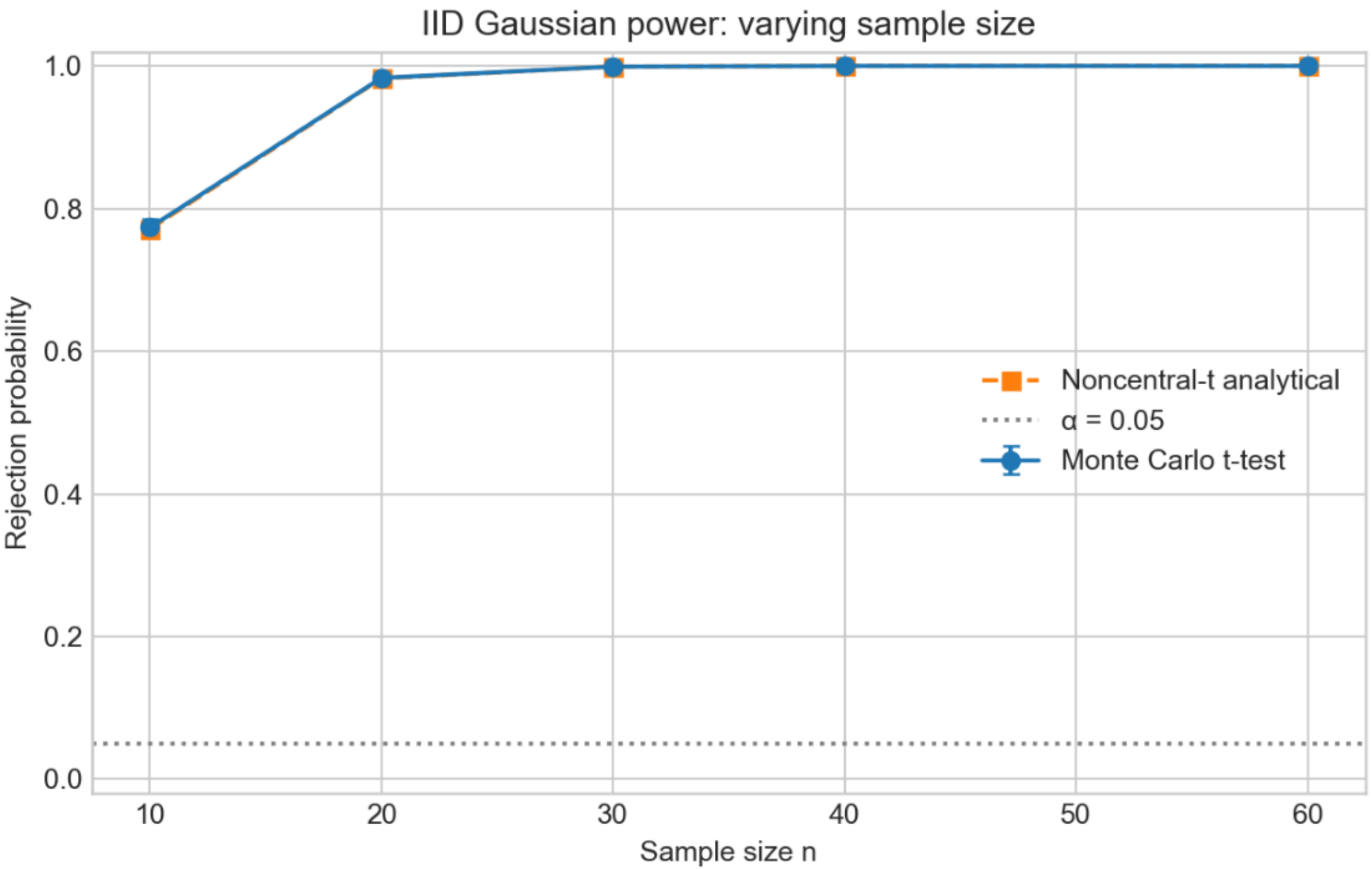

Observed NOAA daily paired-difference distribution
Dates
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−8
−6
−4
−2
0
2
4
6
Night minus day (dB)

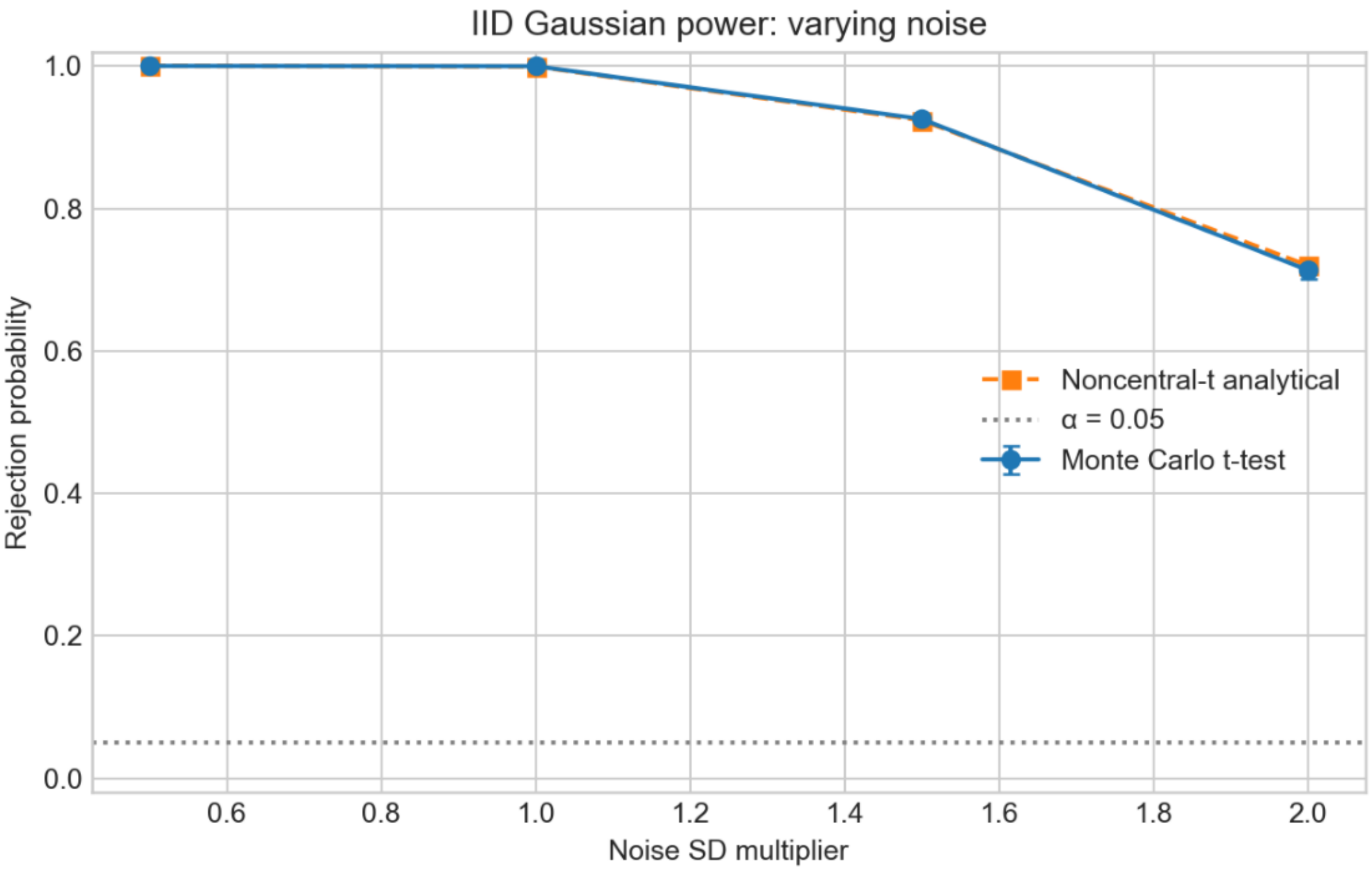

Observed NOAA daily-difference sample ACF
Approx. white-noise band
Autocorrelation
0.2
0.1
0.0
−0.1
−0.2
0
2
4
6
8
10
12
14
Lag (days)

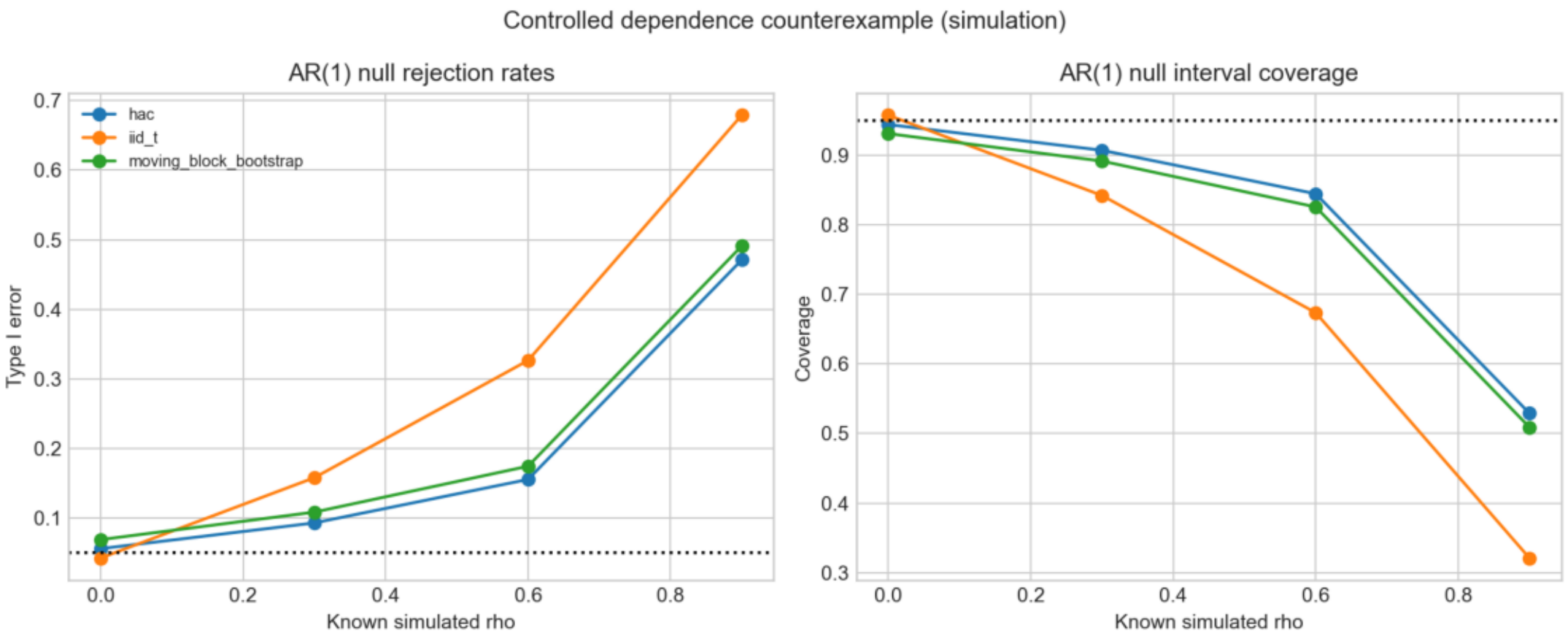

Observed NOAA estimate and 95% intervals
Moving block bootstrap L=8
Moving block bootstrap L=4
Moving block bootstrap L=2
Newey-West HAC
IID one-sample t
1 dB threshold
3 dB threshold
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Night minus day (dB)


IID Gaussian sampling distributions (simulation)
n=20, Δ=0 dB
n=20, Δ=3 dB
n=30, Δ=0 dB
n=30, Δ=3 dB
n=64, Δ=0 dB
n=64, Δ=3 dB
Density
Simulated sample mean


IID Gaussian power: varying effect
Noncentral-t analytical
α = 0.05
Monte Carlo t-test
Rejection probability
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
True effect Δ (dB)


IID Gaussian power: varying sample size
Rejection probability
1.0
0.8
0.6
0.4
0.2
0.0
10
20
30
40
50
60
Sample size n
Noncentral-t analytical
α = 0.05
Monte Carlo t-test


IID Gaussian power: varying noise
Rejection probability
1.0
0.8
0.6
0.4
0.2
0.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Noise SD multiplier
Noncentral-t analytical
α = 0.05
Monte Carlo t-test


Controlled dependence counterexample (simulation)
AR(1) null rejection rates
hac
iid_t
moving_block_bootstrap
Type I error
Known simulated rho
AR(1) null interval coverage
Coverage
Known simulated rho